

Chapter 20: Treatment-Confounder Feedback

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Chapter 19 identified sequential exchangeability as a key condition for identifying the causal effects of time-varying treatments. Suppose a study satisfies the strongest form of sequential exchangeability, so the measured time-varying confounders suffice to estimate the effect of any treatment strategy. Which confounding adjustment method should we then use? The answer exposes a central problem for time-varying treatments: **treatment-confounder feedback**. When it is present, traditional adjustment methods can be biased even though all the information needed for valid estimation is available. This chapter describes the structure of treatment-confounder feedback and explains why traditional methods fail.

This chapter is based on Hernán and Robins (2020, chap. 20, pp. 267-275).

Key insight: with treatment-confounder feedback, a time-varying confounder L_1 is also affected by (or shares causes with) earlier treatment A_0 . Stratifying on L_1 removes confounding for the later treatment A_1 but opens a collider path that creates selection bias for A_0 . Not adjusting for L_1 leaves A_1 confounded. Either way the estimate of a treatment-strategy effect is biased; g-methods (Chapter 21) avoid both problems.

1 20.1 The Elements of Treatment-Confounder Feedback (pp. 267-269)

Return to the sequentially randomized HIV trial of Chapter 19:

- treatment A_k (1: treated, 0: untreated) at each month $k = 0, 1, \dots, K$;
- covariates L_k (CD4 cell count) at each month;
- an outcome Y measuring health status at month $K + 1$.

The time-varying covariates L_k are time-varying confounders: CD4 count L_k affects treatment A_k . In addition, treatment A_{k-1} raises future CD4 count L_k .

Definition 1.1 (Treatment-Confounder Feedback). There is **treatment-confounder feedback** when the time-varying confounder affects treatment ($L_k \rightarrow A_k$) *and* earlier treatment affects the confounder ($A_{k-1} \rightarrow L_k$): “the confounder affects the treatment and the treatment affects the confounder” (Hernán and Robins 2020, 267).

The book’s Figure 20.1 (the same diagram as Figure 19.2) shows the first two months of the trial. As in Chapter 19, the example has no censoring, so we can focus on confounding.

1.1 Time-Varying Confounding Without Feedback

Time-varying confounding can occur *without* treatment-confounder feedback. In the book’s Figure 20.2, the arrows from A_{k-1} to L_k and to U_k are deleted: L_k still confounds the effect of A_k , but it is not affected by prior treatment. There is time-varying confounding but no feedback.

i Fine Point 20.1: Representing Feedback Cycles with Acyclic Graphs

A feedback loop between treatment and confounder can be drawn on an acyclic graph by unrolling it in time: $A_{k-1} \rightarrow L_k \rightarrow A_k \rightarrow L_{k+1} \rightarrow \dots$. This representation requires discrete time: treatment and covariates may change during each interval $[k, k + 1)$, without specifying when within the interval. In practice the intervals can be made as short as the data’s granularity requires (months in the HIV example, where patients see their doctors at most monthly), and time is usually recorded in discrete units anyway (Hernán and Robins 2020, Fine Point 20.1, p. 268).

1.2 The Simplest Case: Figure 20.3

The smallest diagram that shows treatment-confounder feedback in a two-time-point sequentially randomized trial is the book’s Figure 20.3. Its arrows include, among others,

$$A_0 \rightarrow L_1, \quad L_1 \rightarrow A_1, \quad U_1 \rightarrow L_1, \quad U_1 \rightarrow Y.$$

It makes four simplifications relative to Figure 20.1:

- no baseline node L_0 : A_0 is marginally randomized and A_1 is randomized conditional on L_1 ;
- no unmeasured U_0 ;
- no arrow $A_0 \rightarrow A_1$: treatment at time 1 is assigned using L_1 only;
- no arrows from A_0 , L_1 , or A_1 into Y : the **sharp null** holds.

None of these simplifications affects the arguments of the chapter; a larger diagram would add no conceptual insight and would only be harder to read.

Because Figure 20.3 has no forward-directed path from A_0 or A_1 to Y , the true average causal effect of “always treat” versus “never treat” is zero:

$$\mathbb{E}[Y^{a_0=1, a_1=1}] - \mathbb{E}[Y^{a_0=0, a_1=0}] = 0.$$

There are no arrows from the unmeasured U_1 into the treatments, so Figure 20.3 can represent a sequentially randomized trial (or an observational study with no unmeasured confounding), and Chapter 19's results say the observed data on (A_0, L_1, A_1, Y) should identify this effect as 0.

1.3 Feedback Through Shared Causes: Figure 20.4

The same problem arises when the time-varying confounder is not affected by prior treatment but **shares a cause** W_0 with it (the book's Figure 20.4, a subset of Figure 19.4). Its arrows include, among others,

$$W_0 \rightarrow A_0, \quad W_0 \rightarrow L_1, \quad L_1 \rightarrow A_1, \quad U_1 \rightarrow L_1, \quad U_1 \rightarrow Y.$$

Figure 20.4 represents an observational study. The book refers to both Figures 20.3 and 20.4 (and 19.2 and 19.4) as examples of treatment-confounder feedback.

Remark 1.1 (The Claim of This Chapter). With time-varying confounders and treatment-confounder feedback, traditional methods (stratification, outcome regression, matching) cannot correctly adjust for the confounders, even when the data suffice for sequential exchangeability. G-methods adjust correctly even in the presence of feedback.

2 20.2 The Bias of Traditional Methods (pp. 269-271)

A hypothetical sequentially randomized trial with 32,000 individuals with HIV and two time points $k = 0, 1$ (full adherence, no loss to follow-up):

- $A_0 = 1$ is randomly assigned at baseline with probability 0.5;
- A_1 is randomly assigned at month 1 with probability 0.4 if $L_1 = 0$ (high CD4) and 0.8 if $L_1 = 1$ (low CD4);
- the outcome Y , measured at the end of follow-up, is a health score (higher is better).

Example 2.1 (Table 20.1).

N	A_0	L_1	A_1	Mean Y
2400	0	0	0	84
1600	0	0	1	84
2400	0	1	0	52
9600	0	1	1	52
4800	1	0	0	76
3200	1	0	1	76
1600	1	1	0	44
6400	1	1	1	44

Source: Hernán and Robins (2020, Table 20.1, p. 269). Each row is a combination of (A_0, L_1, A_1) , with its number of individuals N and $E[Y \mid A_0, L_1, A_1]$.

The design probabilities can be read back from the table. Within $L_1 = 0$: $1600/(2400 + 1600) = 0.4$ and $3200/(4800 + 3200) = 0.4$. Within $L_1 = 1$: $9600/(2400 + 9600) = 0.8$ and $6400/(1600 + 6400) = 0.8$. Also, 16,000 of the 32,000 individuals have $A_0 = 1$, matching probability 0.5.

The identifiability conditions hold by design: there are no confounders for A_0 , L_1 is the only confounder for A_1 (so sequential exchangeability holds given L_1), and positivity holds because no row of the table is empty. If treatment at later times k were also affected by L_k , then L_k would be a time-varying confounder too. Random variability is ignored throughout.

2.1 Each Treatment, Taken Separately, Has No Effect

Effect of A_1 . In every stratum of (A_0, L_1) , the mean outcome is the same for $A_1 = 1$ and $A_1 = 0$ (rows 1 vs 2, 3 vs 4, 5 vs 6, 7 vs 8). For example,

$$E[Y \mid A_0 = 0, L_1 = 0, A_1 = 1] - E[Y \mid A_0 = 0, L_1 = 0, A_1 = 0] = 84 - 84 = 0,$$

and because the identifiability conditions hold this equals $E[Y^{a_1=1} \mid A_0 = 0, L_1 = 0] - E[Y^{a_1=0} \mid A_0 = 0, L_1 = 0]$.

Effect of A_0 . Because A_0 is randomized, $E[Y \mid A_0 = 1] - E[Y \mid A_0 = 0]$ estimates $E[Y^{a_0=1}] - E[Y^{a_0=0}]$. Averaging rows 1-4 (16,000 individuals with $A_0 = 0$):

$$\begin{aligned} E[Y \mid A_0 = 0] &= \frac{2400 \times 84 + 1600 \times 84 + 2400 \times 52 + 9600 \times 52}{16000} \\ &= \frac{4000 \times 84 + 12000 \times 52}{16000} = \frac{336000 + 624000}{16000} = 60. \end{aligned}$$

Averaging rows 5-8 (16,000 individuals with $A_0 = 1$):

$$\begin{aligned} E[Y \mid A_0 = 1] &= \frac{4800 \times 76 + 3200 \times 76 + 1600 \times 44 + 6400 \times 44}{16000} \\ &= \frac{8000 \times 76 + 8000 \times 44}{16000} = \frac{608000 + 352000}{16000} = 60. \end{aligned}$$

So the average causal effect of A_0 is $60 - 60 = 0$.

i Technical Point 20.1: G-Null Test

Under the sharp null, every counterfactual outcome Y^g equals the observed Y . With two time points, sequential exchangeability for all Y^g then reads $Y \perp\!\!\!\perp A_0 \mid L_0$ and $Y \perp\!\!\!\perp A_1 \mid A_0, L_0, L_1$ (using the fact that for any a_0 and l_0 some strategy g has $a_0 = g(l_0)$). Under sequential exchangeability, testing these conditional independencies is a test of the sharp null: the **g-null test** (Robins 1986). The first independence says A_0 has no effect within strata of L_0 ; the second says A_1 has no effect within strata of (A_0, L_1) .

More generally, the **g-null theorem** (Robins 1986) states that, under sequential randomization for all g , these two independencies hold if and only if the distribution of Y^g (and therefore $E[Y^g]$) is the same for all g and equal to the distribution (and mean) of the observed Y (Hernán and Robins 2020, Technical Point 20.1, p. 270).

2.2 Analyzing the Joint Treatment

Since the effects of A_0 and A_1 are each null, the g-null theorem implies $E[Y^{a_0=1, a_1=1}] - E[Y^{a_0=0, a_1=0}] = 0$. Do conventional analyses recover this?

Analysis 1: no adjustment for L_1 . Compare the 9600 individuals treated at both times (rows 6 and 8) with the 4800 untreated at both times (rows 1 and 3):

$$\begin{aligned} E[Y \mid A_0 = 1, A_1 = 1] &= \frac{3200 \times 76 + 6400 \times 44}{9600} = \frac{243200 + 281600}{9600} \approx 54.7, \\ E[Y \mid A_0 = 0, A_1 = 0] &= \frac{2400 \times 84 + 2400 \times 52}{4800} = \frac{201600 + 124800}{4800} = 68.0. \end{aligned}$$

The difference $54.7 - 68 = -13.3$ is non-null, because $E[Y \mid A_0 = a_0, A_1 = a_1]$ is not a valid estimator of $E[Y^{a_0, a_1}]$: adjustment for the confounder L_1 is needed.

Analysis 2: stratification on L_1 . Within $L_1 = 0$, compare treated at both times (row 6) with untreated at both times (row 1):

$$E[Y \mid A_0 = 1, L_1 = 0, A_1 = 1] - E[Y \mid A_0 = 0, L_1 = 0, A_1 = 0] = 76 - 84 = -8.$$

Within $L_1 = 1$ (rows 8 and 3): $44 - 52 = -8$.

The analysis adjusted for the confounder also gives the wrong answer.

Because the stratum-specific difference is -8 in *both* strata of L_1 , no weighted average of the stratum-specific differences can equal the correct value 0. This estimate reflects the bias of traditional methods when there is treatment-confounder feedback (Hernán and Robins 2020, 271).

Preview of Chapter 21. The correct answer can be recovered from the same table by the g-formula, which standardizes over the distribution of L_1 *given* A_0 rather than conditioning on L_1 . From Table 20.1, $\Pr[L_1 = 0 \mid A_0 = 1] = 8000/16000 = 0.5$ and $\Pr[L_1 = 0 \mid A_0 = 0] = 4000/16000 = 0.25$, so

$$\begin{aligned} E[Y^{a_0=1, a_1=1}] &= 76 \times 0.5 + 44 \times 0.5 = 38 + 22 = 60, \\ E[Y^{a_0=0, a_1=0}] &= 84 \times 0.25 + 52 \times 0.75 = 21 + 39 = 60, \end{aligned}$$

and the difference is $60 - 60 = 0$, as it should be. Chapter 21 develops this computation.

3 20.3 Why Traditional Methods Fail (pp. 271-273)

All three identifiability conditions hold in Table 20.1; the unmeasured U_1 (immunosuppression level) is not needed because we have data on L_1 . Yet neither analysis gave the correct answer. The problem is the **adjustment method**: stratification cannot handle treatment-confounder feedback.

3.1 Stratification Opens a Collider Path

Stratifying on L_1 means estimating the treatment-outcome association separately in the subsets $L_1 = 0$ (high CD4) and $L_1 = 1$ (low CD4). But L_1 is affected by prior treatment A_0 , so it is a **collider** on the path

$$A_0 \rightarrow L_1 \leftarrow U_1 \rightarrow Y$$

(the book's Figure 20.5: Figure 20.3 with L_1 conditioned on). Conditioning on L_1 opens this path and induces a noncausal association between A_0 and U_1 , and hence between A_0 and Y , within levels of L_1 .

Intuition: among those with low CD4 count ($L_1 = 1$), being on treatment ($A_0 = 1$) marks severe immunosuppression (high U_1), since treatment would otherwise have raised their CD4. Among those with high CD4 count ($L_1 = 0$), being off treatment ($A_0 = 0$) marks milder immunosuppression (low U_1).

Remark 3.1 (Trading Confounding for Selection Bias). Stratification eliminates confounding for A_1 at the cost of introducing selection bias for A_0 . The associational differences

$$E[Y \mid A_0 = 1, L_1 = l, A_1 = 1] - E[Y \mid A_0 = 0, L_1 = l, A_1 = 0]$$

may be non-zero even when treatment has no effect on anyone's outcome at any time. The net

bias depends on the relative sizes of the confounding removed and the selection bias created.

3.2 Feedback Through Shared Causes

The same bias arises when the confounder shares an unmeasured cause W_0 with prior treatment (an observational study, the book's Figure 20.6): conditioning on the collider L_1 opens

$$A_0 \leftarrow W_0 \rightarrow L_1 \leftarrow U_1 \rightarrow Y.$$

The settings of Figures 20.3 and 20.4 cannot be distinguished using the observed data, which is why both are called treatment-confounder feedback.

3.3 Beyond the Null and Beyond Two Time Points

- The bias is not limited to the sharp null. With a non-null effect (the book's Figure 20.7, where $A_0 \rightarrow L_1 \rightarrow Y$), arrows from A_0 , A_1 , or L_1 into Y do not change the fact that conditioning on L_1 creates a noncausal association between A_0 and Y .
- With many time points and variables, the bias still arises; confounders affected by prior treatment at multiple times increase the possibility of a large bias.
- Valid estimation of a treatment-strategy effect requires estimating the joint effect of all treatment components A_k simultaneously and without bias, which stratification may be unable to do even with data on all time-varying confounders.

i Fine Point 20.2: Confounders on the Causal Pathway

Conditioning on a confounder L_1 affected by prior treatment can create selection bias even when L_1 is not on a causal pathway from treatment to outcome (no such pathway exists in Figures 20.5 and 20.6).

In Figure 20.7, by contrast, L_1 is a confounder for A_1 *and* lies on the causal pathway $A_0 \rightarrow L_1 \rightarrow Y$. Even if U_1 were not a common cause of L_1 and Y (no selection bias), the A - Y associations within strata of L_1 would estimate only the direct effect of A_0 not through L_1 , not the overall effect of $\bar{A}$ on Y .

The common rule that variables on a causal pathway cannot be confounders is inaccurate for time-varying treatments: a confounder for later treatment A_1 can lie on a pathway from earlier treatment A_0 to Y . Whether adjusting for it induces bias depends on the method: stratification does; g-methods do not (Hernán and Robins 2020, Fine Point 20.2, p. 272).

4 20.4 Why Traditional Methods Cannot Be Fixed (pp. 273-274)

Could parametric **outcome regression** succeed where nonparametric stratification failed? The question matters most with high-dimensional data, where a simple stratified analysis is impossible.

- With two time points there are $2^2 = 4$ static strategies $\bar{a}$.
- With K of the order of 100 (not unusual in practice), there are 2^{100} static strategies, far more than the sample size of any study, and even more once dynamic strategies are considered.

The number of combinations of the *data* is larger still, because several confounders L_k are measured at each time point.

4.1 Modeling Is Needed, but Does Not Help

As argued since Chapter 11, many possible strategies require **modeling**: a dose-response function for the effect of treatment history $\bar{a}$ on the mean outcome. For example, assume the effect is linear in **cumulative treatment**, so every strategy with exactly three months of treatment has the same effect, whenever those three months occur. The price is a new threat to validity: misspecification of the dose-response model.

Paying that price buys no protection. Regression is a conventional stratification-based method, so it cannot remove the bias of stratification under treatment-confounder feedback.

4.2 Example: A Cumulative-Treatment Regression

For data generated under Figure 20.5, define $\text{cum}(\bar{A}) = A_0 + A_1 \in \{0, 1, 2\}$. “Always treat” is $\text{cum}(\bar{a}) = 2$ and “never treat” is $\text{cum}(\bar{a}) = 0$; the target is $E[Y^{\text{cum}(\bar{a})=2}] - E[Y^{\text{cum}(\bar{a})=0}]$, whose true value is 0.

Fit the outcome regression model

$$E[Y \mid \bar{A}, L_1] = \theta_0 + \theta_1 \text{cum}(\bar{A}) + \theta_2 L_1.$$

Within either level l of L_1 ,

$$\begin{aligned} E[Y \mid \text{cum}(\bar{A}) = 2, L_1 = l] - E[Y \mid \text{cum}(\bar{A}) = 0, L_1 = l] \\ = (\theta_0 + 2\theta_1 + \theta_2 l) - (\theta_0 + 0 \cdot \theta_1 + \theta_2 l) = 2\theta_1. \end{aligned}$$

It is tempting to read $2\theta_1$ as the effect of “always treat” versus “never treat” within levels of L_1 . But conditioning on L_1 induces an association between A_0 (a component of $\text{cum}(\bar{A})$) and Y , so $\theta_1 \neq 0$ even though the true effect is zero (Hernán and Robins 2020, 274). The bias comes from conditioning on the collider L_1 , not from the form of the model or from the sample size: it would persist with unlimited data.

A similar argument applies to **matching**. G-methods are needed to adjust appropriately for time-varying confounders in the presence of treatment-confounder feedback (Hernán and Robins 2020, 274).

The book states that θ_1 is non-zero “even if the true causal effect is zero and the regression model for $E[Y \mid \bar{A}, L_1]$ is correct” (Hernán and Robins 2020, 274). Under Figure 20.5, the cumulative-treatment model itself cannot be the correct one. Y is independent of A_1 given (A_0, L_1) , so a correct model for $E[Y \mid A_0, A_1, L_1]$ has no A_1 term, whereas $\text{cum}(\bar{A})$ forces A_0 and A_1 to share the coefficient θ_1 . The data in Table 20.1 illustrate this. Their cell means depend on (A_0, L_1) only, and they are fit exactly by

$$E[Y \mid A_0, A_1, L_1] = 84 - 8A_0 + 0 \cdot A_1 - 32L_1,$$

which reproduces all four distinct cell means:

- $A_0 = 0, L_1 = 0$: 84;
- $A_0 = 0, L_1 = 1$: $84 - 32 = 52$;
- $A_0 = 1, L_1 = 0$: $84 - 8 = 76$;
- $A_0 = 1, L_1 = 1$: $84 - 8 - 32 = 44$.

The conclusion is unchanged: even this correctly specified model gives A_0 a non-zero coefficient (-8) although the true effect of A_0 is zero, because the model conditions on the collider L_1 .

5 20.5 Adjusting for Past Treatment (pp. 274-275)

So far the diagrams had no arrow $A_0 \rightarrow A_1$. Now suppose doctors use past treatment history $\bar{A}_{k-1}$ when deciding on treatment A_k . Adding $A_0 \rightarrow A_1$ to Figures 20.3 and 20.4 gives the book's Figures 20.8 and 20.9.

With treatment-confounder feedback, conditioning on L_1 alone no longer blocks all backdoor paths between A_1 and Y ; it opens

$$A_1 \leftarrow A_0 \rightarrow L_1 \leftarrow U_1 \rightarrow Y \quad (\text{Figure 20.8}),$$

$$A_1 \leftarrow A_0 \leftarrow W_0 \rightarrow L_1 \leftarrow U_1 \rightarrow Y \quad (\text{Figure 20.9}).$$

And whenever past treatment affects the outcome (Figure 20.10), A_0 is a confounder for the effect of A_1 , feedback or not.

Remark 5.1 (Sequential Exchangeability Conditions on Treatment History). Sequential exchangeability at time k generally requires conditioning on the treatment history $\bar{A}_{k-1}$ as well as on the covariates; conditioning only on L is not enough. That is why every sequential exchangeability statement in Chapters 19 and 20 conditions on treatment history.

5.1 Past Treatment and Time-Fixed Treatments

Suppose the target is the effect of the time-fixed treatment A_1 alone on Y (sometimes called the **short-term effect** of $\bar{A}$). Without adjustment for A_0 :

- with treatment-confounder feedback, there is selection bias;
- if A_0 affects Y directly, there is confounding.

So $E[Y \mid A_1 = 1, L_1] - E[Y \mid A_1 = 0, L_1]$ need not be zero even if A_1 has no effect on anyone's outcome (Figures 20.8-20.10).

In practice, this bias can arise in analyses comparing current users ($A_1 = 1$) with nonusers ($A_1 = 0$) of a time-fixed treatment. It can be avoided by adjusting for prior treatment history or by restricting to individuals with a particular treatment history. This restriction is the idea behind **new-user designs**: restrict the analysis to individuals with no past use of the treatment. If past treatment could be correctly adjusted for, the restriction to new users would be unnecessary.

5.2 Mismeasured Past Treatment

The need to adjust for past treatment matters when past treatment is **mismeasured**. As in Section 9.3, adjusting for a mismeasured confounder can leave bias in either direction.

Example 5.1 (Self-Reported Treatment History). Suppose HIV investigators lack medical records and ascertain prior treatment by questionnaire, so they observe a mismeasured A_0^* rather than A_0 . Adding the arrow $A_0 \rightarrow A_0^*$ to Figures 20.8-20.10 shows that conditioning on A_0^* cannot block the biasing paths between A_1 and Y that pass through A_0 . Investigators would find an association between A_1 and Y even after adjusting for A_0^* and L_1 , although A_1 has no effect on Y .

Contrary to a widespread belief, mismeasurement of a time-varying treatment (even independent, non-differential measurement error) **may cause bias under the null**, because past treatment confounds the effect of later treatment even when past treatment has no effect on the outcome. Under the alternative, the imperfect adjustment may exaggerate the effect estimate. Robins (1987)

showed that randomly mismeasured treatment may lead to bias away from the null (Hernán and Robins 2020, 275).

6 Summary

- **Treatment-confounder feedback:** a time-varying confounder L_k affects later treatment A_k and is affected by (or shares causes with) earlier treatment.
- In Table 20.1 the true effect of “always treat” vs. “never treat” is 0, yet the unadjusted analysis gives -13.3 and stratification on L_1 gives -8 .
- Stratification on L_1 removes confounding for A_1 but opens the collider path $A_0 \rightarrow L_1 \leftarrow U_1 \rightarrow Y$, creating selection bias for A_0 .
- Outcome regression and matching are stratification-based too, so they are biased even when the regression model is correct.
- Sequential exchangeability generally requires conditioning on past treatment; mismeasured past treatment can bias estimates even under the null.
- G-methods (Chapter 21) handle treatment-confounder feedback correctly.

7 References

Hernán, Miguel A, and James M Robins. 2020. *Causal Inference: What If*. Chapman & Hall/CRC. <https://miguelhernan.org/whatifbook>.